Why study limits?

1. To determine PODS

2. To find vertical asymptotes

3. To find horizontal asymptotes

4. To determine Continuity

5. The Squeeze Theorem

Given: $g(x) = \dfrac{x^2 + x - 6}{2x^2 - 2x - 4}$

Calculate each limit and state what that limit tells you about the graph of g(x).

1) $\lim_{x \to -3} g(x)$

2) $\lim_{x \to -1} g(x)$

3) $\lim_{x \to 2} g(x)$

4) $\lim_{x \to \infty} g(x)$

5) $\lim_{x \to -\infty} g(x)$

Given: $g(x) = \dfrac{x^2 + x - 6}{2x^2 - 2x - 4}$

Calculate each limit and state what that limit tells you about the graph of g(x).

1) $\lim_{x \to -3} g(x) = 0$

2) $\lim_{x \to -1} g(x) = \text{DNE}$

3) $\lim_{x \to 2} g(x) = 5/6$

4) $\lim_{x \to \infty} g(x) = 1/2$

5) $\lim_{x \to -\infty} g(x) = 1/2$

Find all POD's and Asymptotes:

$$f(x)=\frac{x^2+x-2}{x^2+3x-4}$$

When is a function continuous at x = a ?

A function $f(x)$ is continuous at $\boldsymbol{x} = a$ if

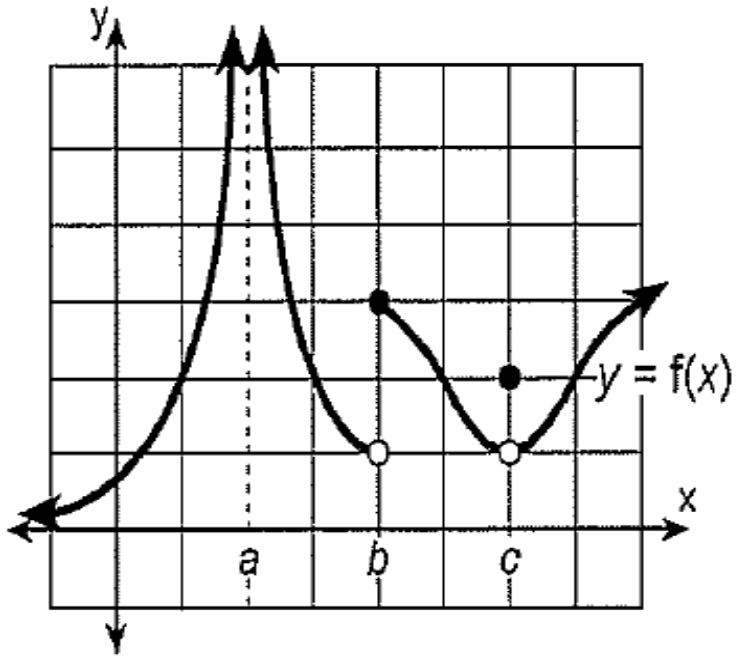
(i) $f(a)$ exists

(ii) $\lim_{x \to a} f(x)$ *exists*

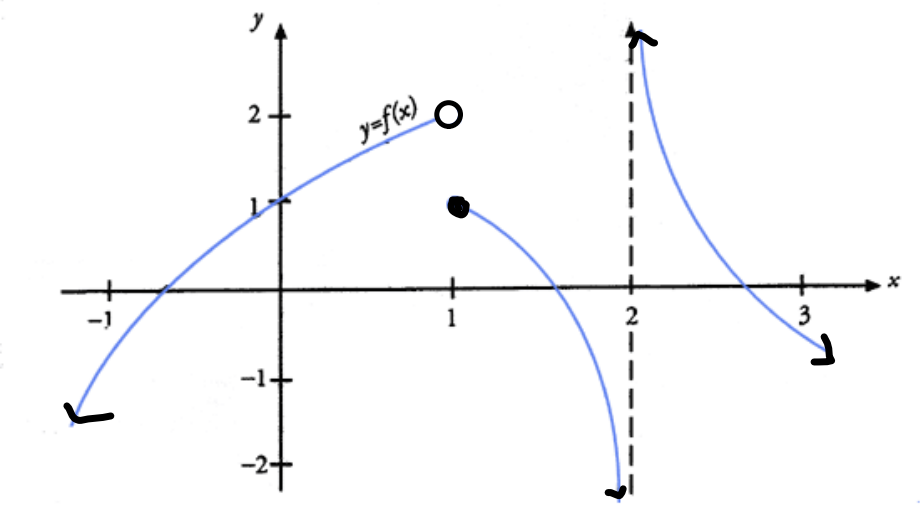
(iii) $\lim_{x \to a} f(x) = f(a)$

or when is a function NOT continuous at x = a ?

Types of discontinuity:



State all values of x for which f(x) is continuous.



For what values of f(x) NOT continuous?

How does continuous work with piecewise functions?

Is *g(x)* continuous at all values of *x*?

$$g(x) = \begin{cases} x^3 + 1, & x \le 1 \\ 2x, & x > 1 \end{cases}$$

The function f is defined below:

$$f(x)=\begin{cases}4x-11, x<3\\ kx^2, x\geq 3\end{cases}$$

Find a value of k such that f is continuous at all values of x.

Fix a discontinuity....

Find the value of k for which $f(x)$ is continuous for all values of x.

$$f(x) = \begin{cases} \dfrac{\sin 3x}{x}, x \neq 0 \\ k, \; x = 0 \end{cases}$$

Find a value of c that makes the function $f(x)$ below continuous at $x = 0$.

$$f(x) = \begin{cases} \dfrac{1-\cos 3x}{x^2}, if\ x \neq 0 \\ c, if\ x = 0 \end{cases}$$

If $h(x) = \begin{cases} 3x^2 - 2 & , \ x < -1 \\ f(x) & , \ x \geq -1 \end{cases}$, then $h(x)$ will

be a continuous function at $x = -1$ when $f(x) =$

A) $x^3 - 1$ B) $x + 1$ C) 2^{x-1}

D) $\sin(x + 1)$ E) $\cos(x + 1)$

Given $f(x) = \begin{cases} \dfrac{2x^2 - 6x}{x^2 - x - 6}, & x \neq -2, 3 \\ \dfrac{6}{5}, & x = 3 \\ 0, & x = -2 \end{cases}$.

Which of the following statements is true?

A) Function f is continuous, except at $x = 3$.

C) Function f is continuous, except at $x = 0$ or 3.

E) Function f is continuous, except at $x = -2$ or 3.

B) Function f is continuous, except at $x = -2$, 0, or 3.

D) Function f is continuous, except at $x = -2$.